

Math 421

Wednesday, October 29

Announcements

- Grace's Drop-In Hours
 - **Today** 12-1 pm VV 311
 - Thursday 9-11 am VV B205
- MLC
 - Proof Table: M-Th 3-7:30 VV B227
 - Course Assistant: Th 4-7 Table 4
- Homework 7 Due Friday
- Exam 2 – 11/12 5:45-7:15 pm
 - Email me & fill out the form if you have a conflict and need to take the makeup.
 - Sign up for McBurney exam if applicable.

$$a \leq b + \varepsilon \quad \forall \varepsilon > 0 \\ \Rightarrow a \leq b$$

$(\Rightarrow)$ Lemma 10.9

$$\liminf s_n \leq \limsup s_n$$

• Cauchy $\Rightarrow$ bounded

• Thm 10.7 *

$\lim s_n$ exists $\Leftrightarrow$

$$\liminf s_n = \limsup s_n$$

Want to prove: $\limsup s_n \leq \liminf s_n$.

Convergent $\Leftrightarrow$ Cauchy

Theorem 10.11. A sequence is a convergent sequence if and only if it is a Cauchy sequence.

Proof. (Cauchy $\Rightarrow$ Convergent). Let $\varepsilon > 0$. Since (s_n) is a Cauchy sequence then there exists N such that $m, n > N$
 $|s_n - s_m| < \varepsilon$. $-\varepsilon < s_n - s_m < \varepsilon$ so $s_n < s_m + \varepsilon$. For all $m, n > N$
 $s_m + \varepsilon$ is an upper bound $\{s_n : n > N\}$ and $\sup\{s_n : n > N\} \leq s_m + \varepsilon$
 for all $m, n > N$. So $\forall_N - \varepsilon \leq s_m$ for all $m > N$. Therefore
 $u_N = \inf\{s_m : m > N\} \geq v_N - \varepsilon$. and $v_N \leq u_N + \varepsilon$.

$$\limsup s_n = \lim v_N \leq v_N \leq u_N + \varepsilon \leq \lim u_N + \varepsilon = \liminf s_n + \varepsilon$$

$\swarrow$ decreasing sequence
 $\nwarrow$ increasing sequence

So $\limsup s_n \leq \liminf s_n$. Therefore, since $\liminf s_n \leq \limsup s_n$,
 $\limsup s_n = \liminf s_n = \lim s_n = S \in \mathbb{R}$ because (s_n) is bounded.

$$V_N = \sup \{ s_n : n > N \}$$

$$V_{N+1} = \sup \{ s_n : n > N+1 \}$$

Chapter 2: Sequences

Section 14: Series

Cauchy Criterion

$$s_n = \sum_{k=1}^n a_k$$

$$(s_n) = \left(\sum_{k=1}^1 a_k, \sum_{k=1}^2 a_k, \sum_{k=1}^3 a_k, \dots, \sum_{k=1}^n a_k, \dots \right)$$

Def. We say a series $\sum_{n=1}^{\infty} a_n$ satisfies the Cauchy criterion if its sequence of partial sums is a Cauchy sequence.

(s_n) for all $\varepsilon > 0$, there exists N such that $n, m > N, |s_n - s_m| < \varepsilon$
 or $\left| \sum_{k=1}^n a_k - \sum_{k=1}^m a_k \right| = \left| \sum_{k=m+1}^n a_k \right| < \varepsilon$.

★ for all $\varepsilon > 0$, there exists N such that $n \geq m > N$ implies

$$\left| \sum_{k=m}^n a_k \right| < \varepsilon.$$

→ if its seq. of partial sums converges.

Theorem 14.4. A series converges if and only if it satisfies the Cauchy criterion.

seq. of partial sum is Cauchy $\iff$ converges.

Comparison Test

Theorem 14.6 (i). Let $\sum a_n$ be a series where $a_n \geq 0$ for all n . If $\sum a_n$ converges and $|b_n| \leq a_n$ for all n , then $\sum b_n$ converges.

Proof. Sketch.

let $\epsilon > 0$, there exists N such that $n, m > N$

$$\left| \sum_{k=m}^n a_k \right| < \epsilon$$

$$\left| \sum_{k=m}^n b_k \right| \leq \sum_{k=m}^n |b_k| \leq \sum_{k=m}^n a_k = \left| \sum_{k=m}^n a_k \right| < \epsilon$$